BM4Midpoint: Arithmetic Diagonal Projection

GC2D numerical methods

Explicit palindromic operations

All ABBA and BM4 presets inherit `PalindromicMethod`. For signed $d = ch$, the elementary operations are

$$A_d(u, v, t) = (u + df(t, v), v, t), \qquad B_d(u, v, t) = (u, v + df(t, u), t), \qquad T_d(u, v, t) = (u, v, t + d).$$

Only T advances time. A coefficient $c = 1/2$ means duration $h/2$; there is no additional half inside A or B. The optional exact C flow rotates the copy difference and requires an independent coupling object. Both BM4 presets omit C.

In chronological order, standard ABBA2 is $A_{h/2}, B_{h/2}, T_h, B_{h/2}, A_{h/2}$. The distinct symmetric variant $T_{h/2}, A_{h/2}, B_{h/2}, B_{h/2}, A_{h/2}, T_{h/2}$ evaluates every spatial shear at the midpoint time. Both exhibit second order on the tested nonautonomous rotation. Arbitrary palindromes do not inherit fourth or sixth order, and an arithmetic projection need not be reversible or symplectic.

`Composition` denotes an explicit tuple of operations. Consecutive time translations combine as $T_a T_b = T_{a+b}$. The generic `PalindromicImplicit` and `PalindromicMidpoint` methods select one exterior Hairer projection or a final arithmetic mean. The signed spatial flows and passive momentum normalization are retained; observers reconstruct their groups from elementary traces without replay. NumPy and JAX execute the same explicit operations and differentiate their retained shear inputs. See `docs/models/palindromic/simulation/palindromic-operations.md`.

Current uncoupled configuration

Both `BM4Implicit` and `BM4Midpoint` omit the harmonic coupling flow and no longer accept a coupling-frequency constructor argument. Their twelve coefficients, signed clock and projection placement are unchanged. The independent operation `HarmonicCoupling` retains C for future compositions. Existing observer/archive rates are fixed at zero for new runs; nonzero historical calculations require their original frozen implementation.

1 Complete-cycle definition

For $\dot{z} = f(t, z)$, embed the accepted state as $Ez = (z, z)$ and define $P(u, v) = (u + v)/2$. One step is

$$z_{n+1} = P\Psi_h(Ez_n),$$

where Ψ_h is the same unprojected twelve-stage BM4 composition used by BM4Implicit. No projection is applied between stages and no multiplier is solved. "Midpoint" refers to the final arithmetic mean.

The coefficient vector is $(b_1, \ldots, b_6, b_6, \ldots, b_1)$, where

$$b_1 = 0.0792036964311957, \qquad b_2 = 0.1303114101821663,$$
$$b_3 = 0.2228614958676077, \qquad b_4 = -0.3667132690474257,$$
$$b_5 = 0.3246481886897062, \qquad b_6 = 0.1096884778767498.$$

The sum is one to floating-point accuracy. Starting with the adjoint, stages alternate adjoint and direct maps with signed duration $d_j = b_j h$. If $s_j = t_n + \sum_{i<j} d_i$, an adjoint stage uses s_j and a direct stage uses $s_j + d_j$. The negative coefficient also moves the stage clock backwards.

At the chosen evaluation time, let

$$A_d(u, v) = (u + df(t, v), v), \qquad B_d(u, v) = (u, v + df(t, u)).$$

The spatial direct map is $A_d \circ B_d$ and its adjoint is $B_d \circ A_d$. Neither includes harmonic coupling. The time translations remain in the complete direct/adjoint stage as specified above. Each stage evaluates the vector field twice. Thus one complete step uses 24 evaluations and one arithmetic projection.

2 State and energy conventions

Twelve direct/adjoint stages followed by one spatial arithmetic mean. Energy tracking is optional and passive. The two shear contributions in each stage update a summed momentum. Dividing this accumulator by two gives the physical kappa normalization. The common state conventions are specified below.

3 Order and geometric limitations

On the invariant diagonal of the exact extended flow, the two copies follow the original dynamics and the harmonic coupling vanishes. For a smooth field, the fourth-order unprojected composition therefore satisfies

$$\Psi_h(Ez) = E\varphi_h(z) + O(h^5).$$

Since $PE = I$ and P is linear, the projected method has local error $O(h^5)$ and global order four under the usual stability assumptions. The passive energy coordinates do not change the physical map or its order.

Arithmetic projection discards the copy difference. Consequently, symmetry or symplecticity of a base composition alone does not establish exact reversibility or symplecticity of the accepted physical map. This method makes neither claim. The recorded infinity norm $\|u_f - v_f\|_\infty$ measures the discarded separation before projection.

4 Executable checks

The test suite compares the physical map with an independent signed-stage replay and checks fourth-order convergence against an exact rotation and a tightly integrated nonautonomous reference with energy tracking both disabled and enabled. It also checks momentum normalization against an augmented reference ODE, particle batching, persistent observer callbacks, and independence from off-grid output sampling. Nonlinear iterations are absent by construction.

5 Physical state and endpoint energy observations

Accepted states contain one physical vector $Z \in \mathbb{R}^{dN}$ and scalar time. GC vectors have $2N$ entries and FC vectors have $4N$; doubled workspaces are transient. InitialConditions supplies values and layout during preparation only. The numerical base retains functions, metadata, run status and copy count; execution owns the initial snapshot and output schedule. Supplied Jacobians are independent particle blocks (N, d, d); finite differences use dense matrices.

The generic `EnergyObserver` evaluates only

$$H_n = H(t_n, Z_n), \quad H_{n+1} = H(t_{n+1}, Z_{n+1}), \quad (\partial_t H)_n, \quad (\partial_t H)_{n+1}, \quad \Delta H_n = H_{n+1} - H_n.$$

It also evaluates these physical functions at requested saved samples, without consuming internal stages or integrating a momentum coordinate. The derivative is positive, analytic and partial at fixed state. GC differentiates its normalized Hamiltonian potential with `dt=1`; spatial splines and Fourier time factors retain their original scaling. This is not the secant $\Delta H_n/h$. Variation of a time-dependent Hamiltonian is not a numerical conservation error.

Specialized historical generalized-energy studies separately reconstruct $\dot{\kappa} = -\partial_t H$ from retained numerical stages. Their signed weights, physical normalization and archived diagnostics keep their original meaning. They are not part of generic energy observation. Host and JAX execution share the fixed numerical map and physical-only accepted state. JAX evaluates endpoint energy on the device without a momentum accumulator.

6 Shared palindromic implementation

Public method names remain unchanged; both BM4 constructors now omit harmonic coupling. Their code lives in `src/methods/palindromic/`. Public exports from `simulation` remain available; former ABBA and BM4 import adapters have been removed. The shared execution is `advance`, `solve_projection`, `compose`, then explicit $A/B/T/C$ operations. Arithmetic midpoint variants replace the nonlinear projection by one final average.

The composition retains signed shear inputs for exact particle-block tangents, passive energy quadrature, and existing observation records. Newton and Broyden share the spatial projection equations; the energy observer accumulates only the accepted trace, with the existing factor $1/2$ for physical κ. Adjoint stages evaluate at the start of their signed substep, direct stages at its end. The coefficients remain unchanged. ABBA2Implicit, ABBA4Implicit and BM4Implicit each use one outer projection around their complete tuples, and directly inherit `ProjectedPalindromicMethod`. ABBA6 is removed; its saved results retain their historical labels. The internal family name does not enable the retired full time/momentum projection.

The implementation architecture and current diagrams are documented in `docs/models/palindromic/simulation/palindromic-simulation-architecture.md`.

7 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.